

04 – Databases

A Short Introduction to Databases

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OUTLINE

- 1) Data & databases
- 2) The Relational Databases
- 3) The Entity-Relationship (ER) Model
 - 1) The conceptual model
 - 2) The logical model
 - 3) The physical model
- 4) Structured Query Language (SQL)
- 5) SQLite3
- 6) SQLite3, Node and Express

1) DATA & DATABASES

- ” In common usage, data is a collection of discrete or continuous values that convey information, describing the quantity, quality, fact, statistics, other basic units of meaning, or simply sequences of symbols that may be further interpreted formally. ” Source: <https://en.wikipedia.org/wiki/Data>
- Data about me:
 - First name, Last name, Birth place, Date of birth, Age
 - Address, Number, Street, Zip code, Country...
- Data about a car:
 - Brand, Model, Color, Engine power, Energy, Kilometers, Year...

1) DATA & DATABASES

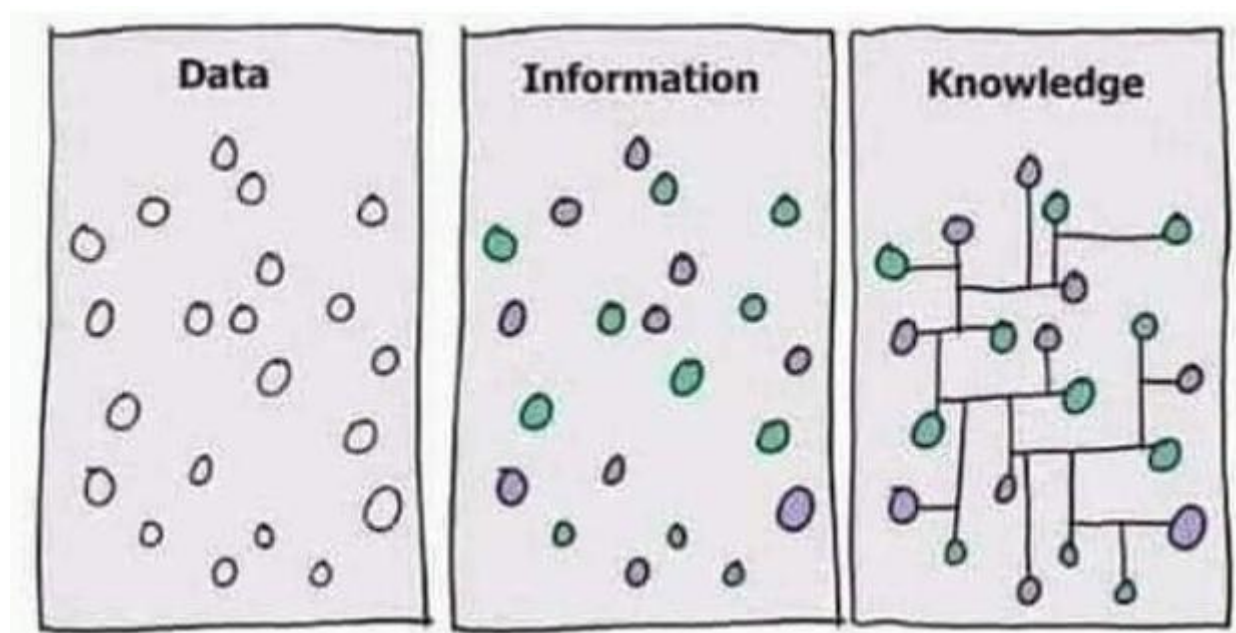
- ” In computing, a database is an organized collection of data or a type of data store based on the use of a database management system (DBMS), the software that interacts with end users, applications, and the database itself to capture and analyze the data. ” Source: <https://en.wikipedia.org/wiki/Database>
- There are many DataBase Management Systems (DBMS)

1) DATA & DATABASES

- There are many models for databases:
 - Navigational DBMS (1960s)
 - Relational DBMS on "big" computers (1970s)
 - Relational DBMS on Personal Computer (1980s)
 - Object Databases and Object-relational Databases (1990s)
 - NoSQL and NewSQL (2000s)
- The classification of databases is complicated, it depends on many parameters. Source: <https://en.wikipedia.org/wiki/Database>

1) DATA & DATABASES

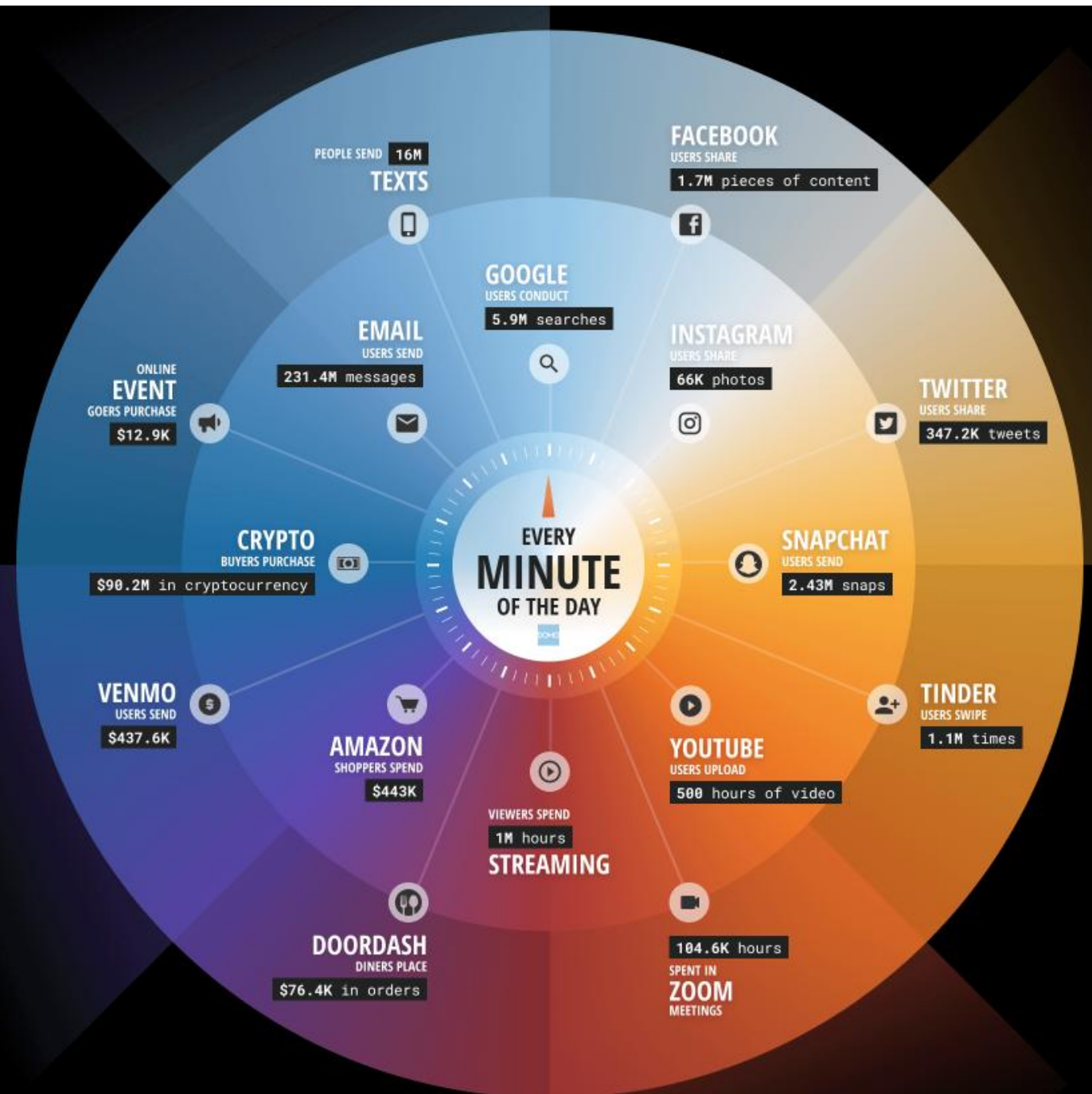
- Difference between Data, Information, and Knowledge... Source: <https://www.linkedin.com/pulse/data-information-knowledge-insights-achin-gupta/>



1) DATA & DATABASES

- Data everywhere...
- Data creation every minute of the day...

Source: <https://www.domo.com/data-never-sleeps>



1) DATA & DATABASES

- Data everywhere...
- Data creation every minute of the day...

Source: <https://www.domo.com/data-never-sleeps>



If you have the data, you have the power!
Data is Money!

2) THE RELATIONAL DATABASES

- ” In a relational database, a relation is a set of tuples that have the same attributes. A tuple usually represents an object and information about that object. Objects are typically physical objects or concepts. A relation is usually described as a table, which is organized into rows and columns. All the data referenced by an attribute are in the same domain and conform to the same constraints. “

Source: https://en.wikipedia.org/wiki/Relational_database

2) THE RELATIONAL DATABASES

- The Relational Databases (RDB) use:
 - **Entities (tables)** containing data organized in rows and columns
 - Columns are the **attributes** or **fields**
 - Rows are the **tuples** or **records**
 - One special column must be the **primary key (PK)** and has a unique value for each tuple in the table

File Edit View Query Project Tools Window Help



Object Explorer

Connect

- External Tables
- Graph Tables
- dbo.AWBuildVersion
- dbo.DatabaseLog
- dbo.ErrorLog
- HumanResources.Department
- HumanResources.Employee
- HumanResources.EmployeeD
- HumanResources.EmployeeP
- HumanResources.JobCandid
- HumanResources.Shift
- Person.Address
- Person.AddressType
- Person.BusinessEntity
- Person.BusinessEntityAddress
- Person.BusinessEntityContact
- Person.ContactType
- Person.CountryRegion
- Person.EmailAddress
- Person.Password
- Person.Person
- Person.PersonPhone
- Person.PhoneNumberType
- Person.StateProvince
- Production.BillOfMaterials
- Production.Culture
- Production.Document
- Production.Illustration
- Production.Location
- Production.Product
- Production.ProductCategory
- Production.ProductCostHist

SQLQuery1.sql - jth...019 (AD\lanjer (69))

```
SELECT TOP (1000) [BusinessEntityID]
, [PersonType]
, [NameStyle]
, [Title]
, [FirstName]
, [MiddleName]
, [LastName]
, [Suffix]
, [EmailPromotion]
, [AdditionalContactInfo]
, [Demographics]
, [rowguid]
, [ModifiedDate]
FROM [AdventureWorks2019].[Person].[Person]
```

110 %

Results Messages

	BusinessEntityID	PersonType	NameStyle	Title	FirstName	MiddleName	LastName	Suffix	EmailPromotion	AdditionalContactInfo	Demographics
1	1	EM	0	NULL	Ken	J	Sánchez	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso.
2	2	EM	0	NULL	Teri	Lee	Duffy	NULL	1	NULL	<IndividualSurvey xmlns="http://schemas.microso.
3	3	EM	0	NULL	Roberto	NULL	Tamburello	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso.
4	4	EM	0	NULL	Rob	NULL	Walters	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso.
5	5	EM	0	Ms.	Gail	A	Erickson	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso.
6	6	EM	0	Mr.	Jossef	H	Goldberg	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso.
7	7	EM	0	NULL	Dylan	A	Miller	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso.
8	8	EM	0	NULL	Diane	L	Margheim	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso.
9	9	EM	0	NULL	Gigi	N	Matthew	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso.
10	10	EM	0	NULL	Michael	NULL	Raheem	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso.
11	11	EM	0	NULL	Ovidiu	V	Cracium	NULL	0	NULL	<IndividualSurvey xmlns="http://schemas.microso.
12	12	EM	0	NULL	Thierry	B	D'Hers	NULL	2	NULL	<IndividualSurvey xmlns="http://schemas.microso.

Query executed successfully.

jthstuds\sql.win.hj.se (15.0 ... | AD\lanjer (69) | AdventureWorks2019 | 00:00:00 | 1 000 rows

3) THE ENTITY-RELATIONSHIP (ER) MODEL



2) THE ENTITY-RELATIONSHIP (ER) MODEL

- To create a Relational Database, the traditional way is to use the **Entity-Relationship (ER) Model** which has three levels:
 - The **conceptual** model: a diagram representing the entities and their relationships
 - The **logical** model: a list of tables
 - The **physical** model: the organization of the information on the computer hard drive (files)

3) THE ENTITY-RELATIONSHIP (ER) MODEL

- Allows you to draw the **database design**
- Is an easy-to-use graphical tool for **modeling data**
- Widely used in database design
- Often done progressively with **iterations** to make it perfect
- Goes from **conceptual** to **logical** to **physical**

Source: Jasmin Jakupovic “Databases” course at JU

3) THE ENTITY-RELATIONSHIP (ER) MODEL

- Contextual data:
 - To set something in context, to further explain, and to give more value: sometimes you need context to a story.

Zoe Foe, 21, F, 070-123 456

- Explained...

Data value	Context
Zoe	First Name
Foe	Last Name
21	Age
F	Gender
070-123 456	Phone

3) THE ENTITY-RELATIONSHIP (ER) MODEL

From data to database

Data value	Context
Zoe	First Name
Foe	Last Name
21	Age
F	Gender
070-123 456	Phone

Database

fName	IName	Age	Gender	Phone
Zoe	Foe	21	F	070-123 456
Erik	Doe	26	M	071-236 456

Entity: Person

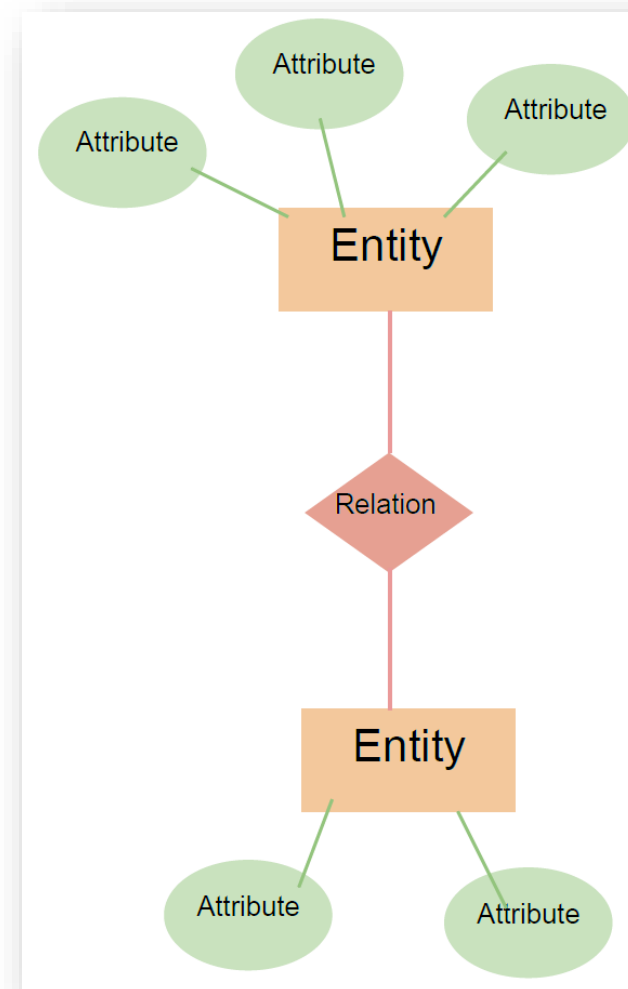
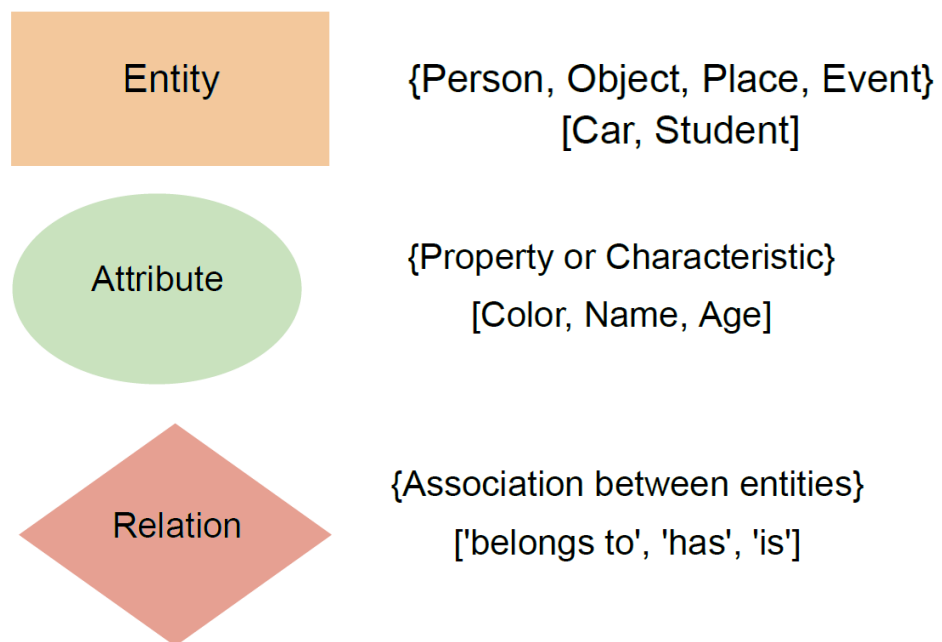
- Context became column (attribute)
- Data value became row (tuple)
- Entity is person
- An 'entity' is a table!

3) THE ENTITY-RELATIONSHIP (ER) MODEL

- Components of the ER Diagram:
 - Entities
 - Attributes
 - Relationships
 - Cardinalities

3.1) THE CONCEPTUAL MODEL

- Notation, how to visualize it!



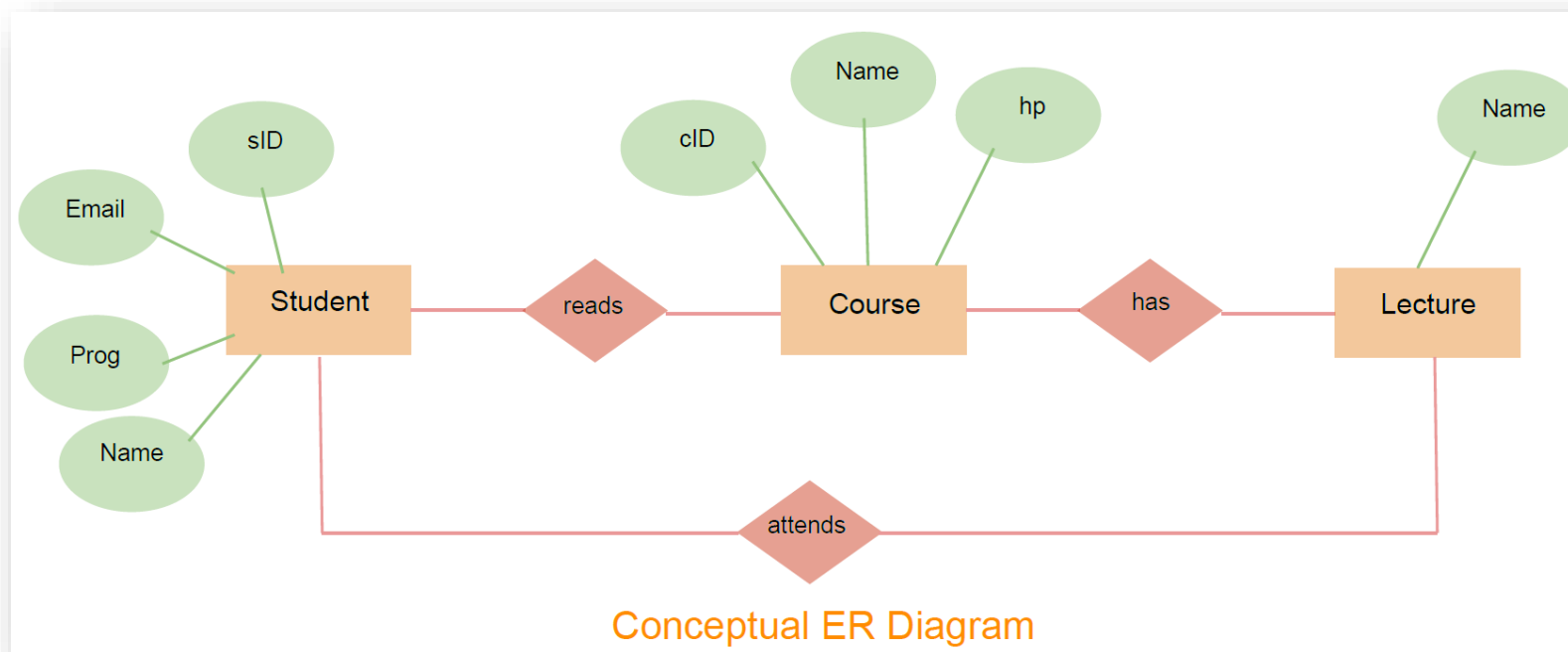
Source: Jasmin Jakupovic "Databases" course at JU

3.1) THE CONCEPTUAL MODEL

- Example: A university database identifies Students, Courses and Lectures as entities. The student has attributes such as ID number, Name, E-mail, Programme. A student usually reads a course which has a number of lectures.

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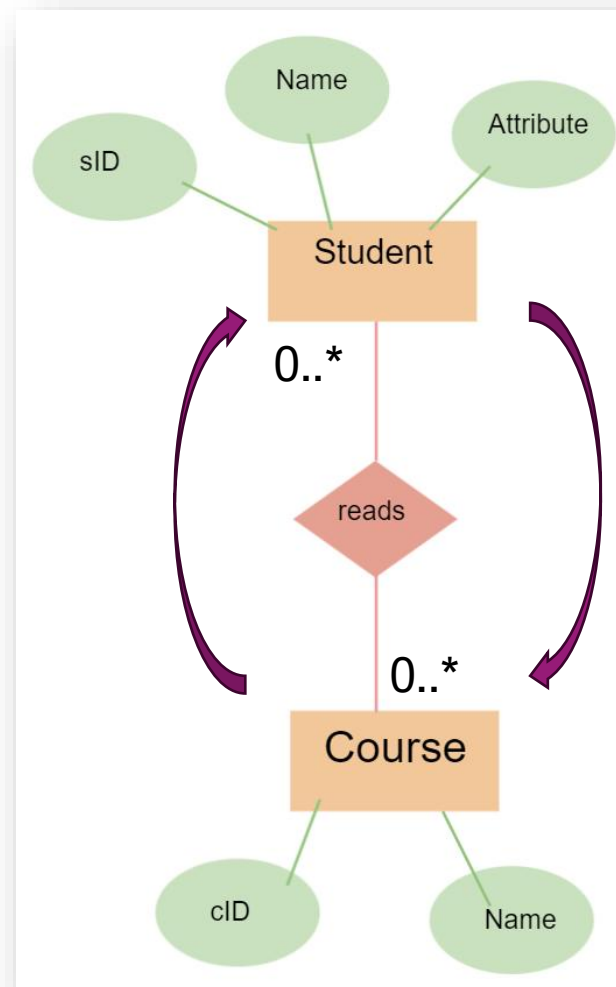
3.1) THE CONCEPTUAL MODEL

- Cardinality:
 - It is a key component of every ER diagram. The cardinality defines the numerical attributes (min..max) of the relationship between two entities or entity sets.

Q1: How many courses does a student have to read?

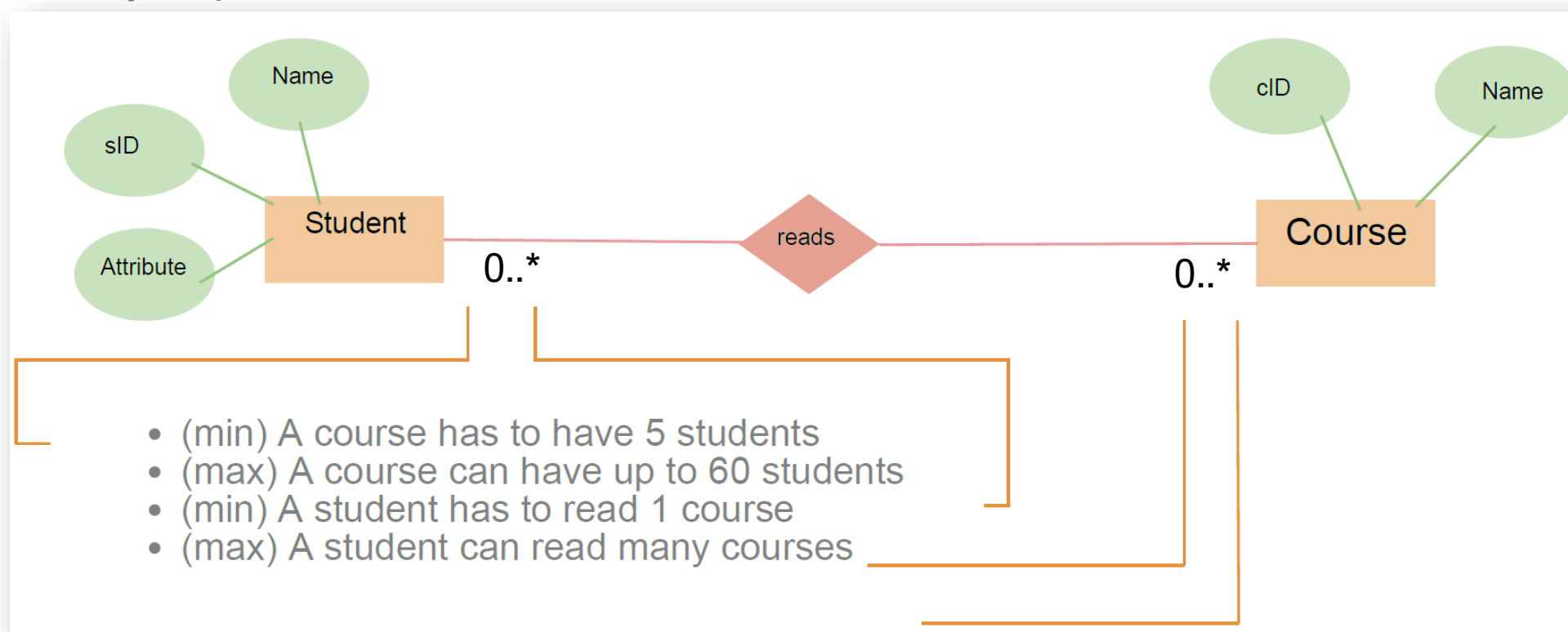
Q2: How many students does a course have?

Source: Jasmin Jakupovic “Databases” course at JU



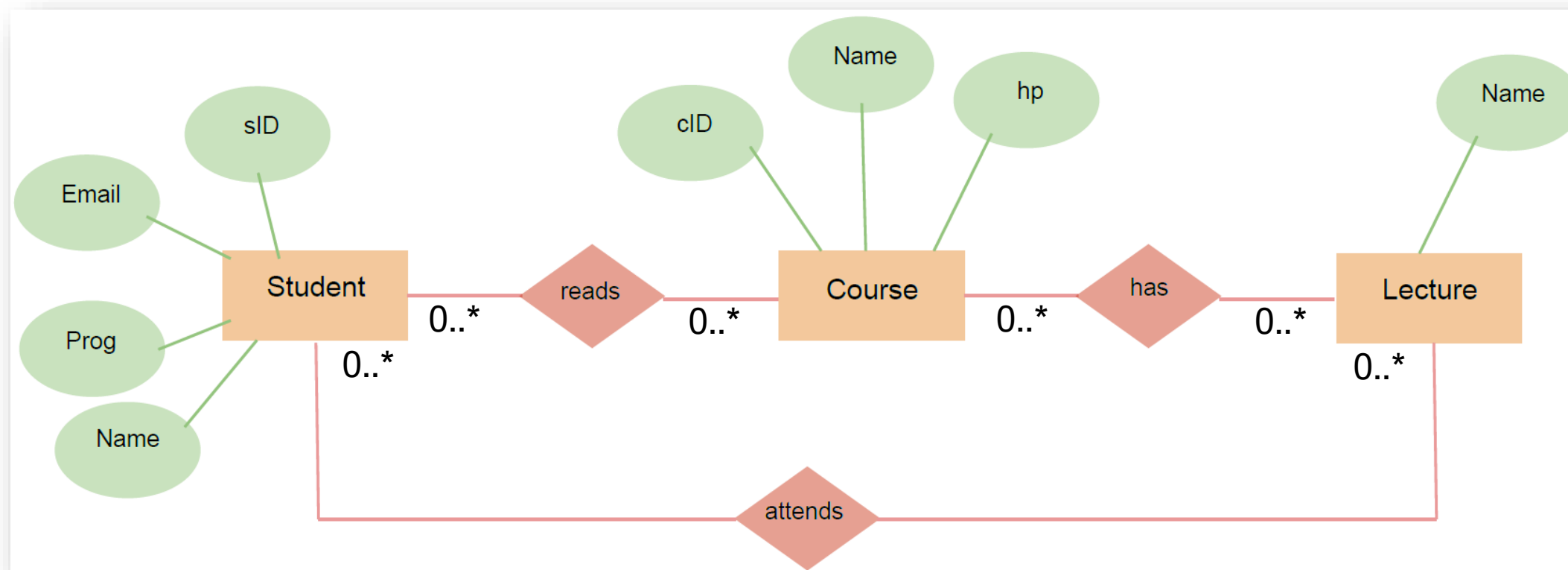
3.1) THE CONCEPTUAL MODEL

- Cardinality explained:



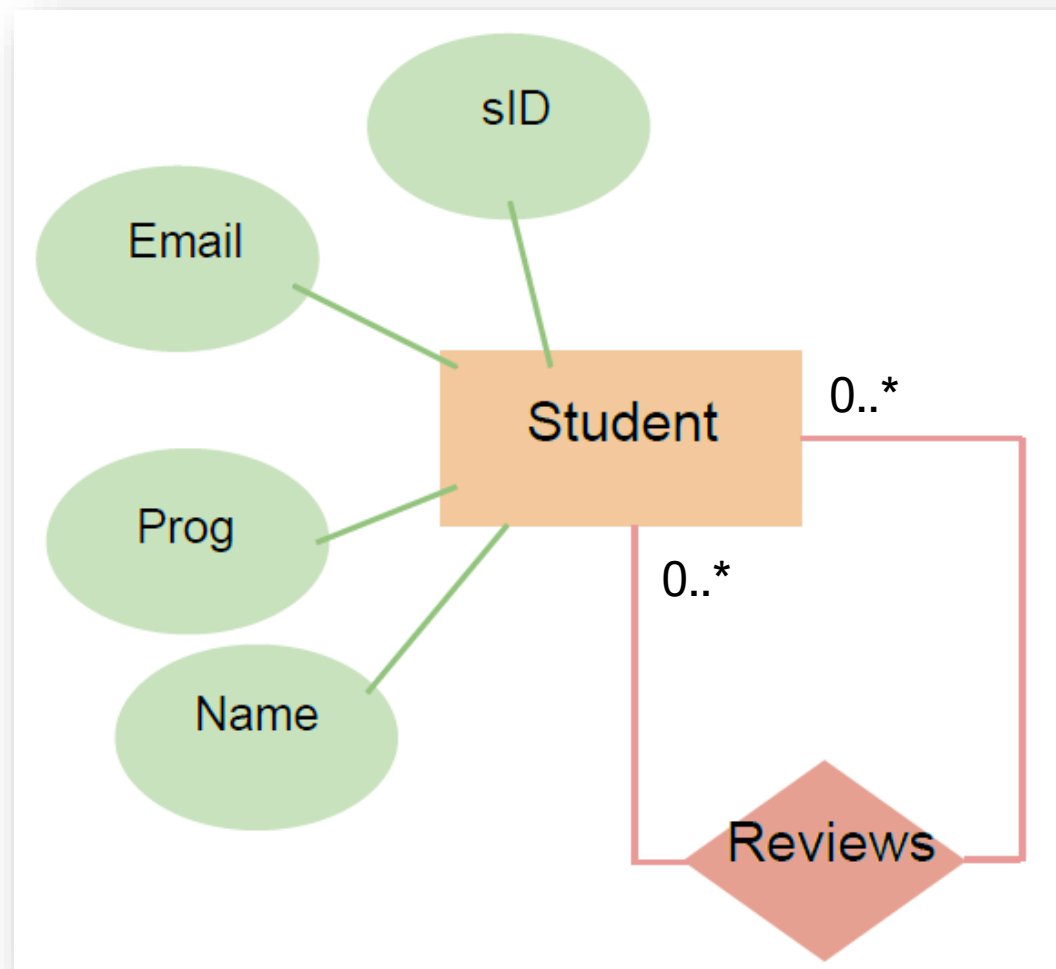
3.1) THE CONCEPTUAL MODEL

- Final conceptual ER diagram with cardinalities:



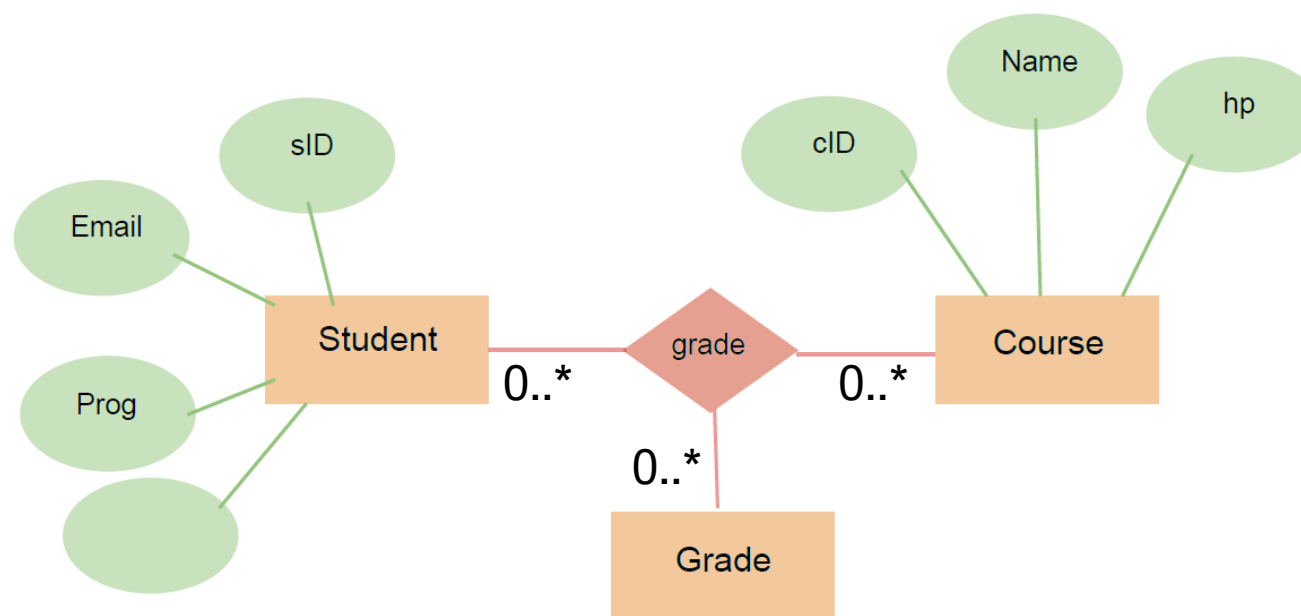
3.1) THE CONCEPTUAL MODEL

- Recursive relations are allowed:



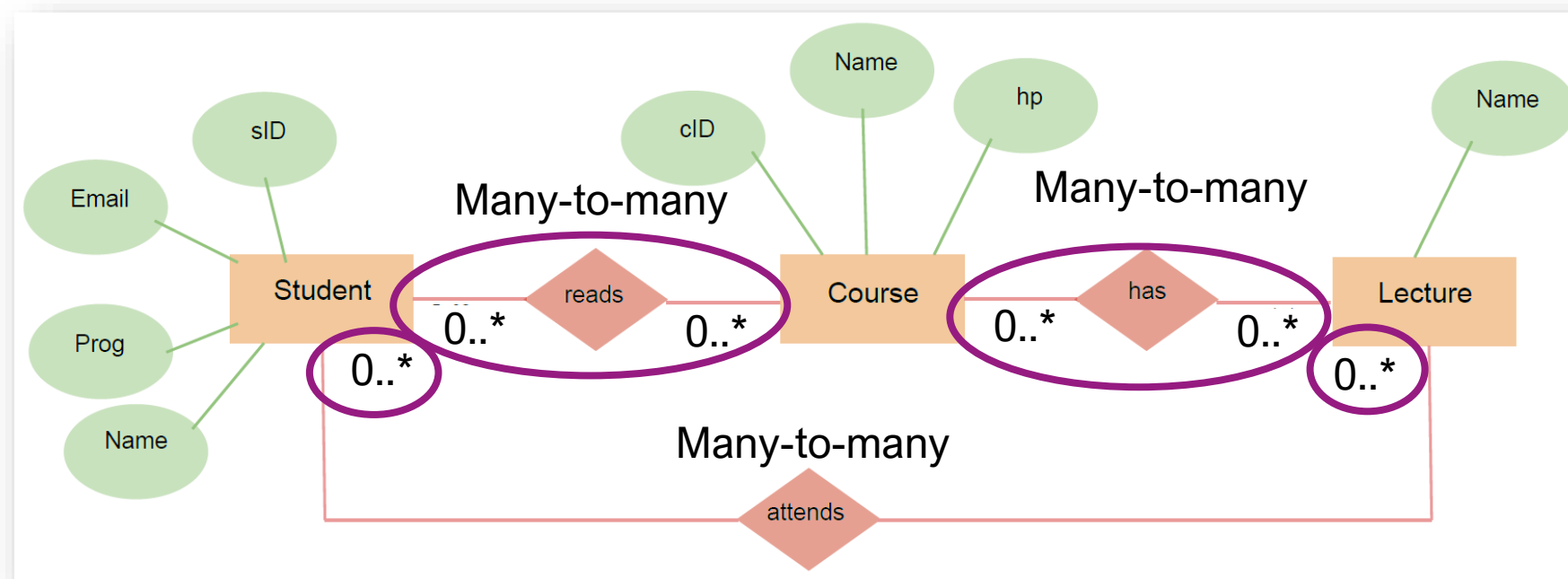
3.1) THE CONCEPTUAL MODEL

- Relations with more than one entities are allowed:



3.1) THE CONCEPTUAL MODEL

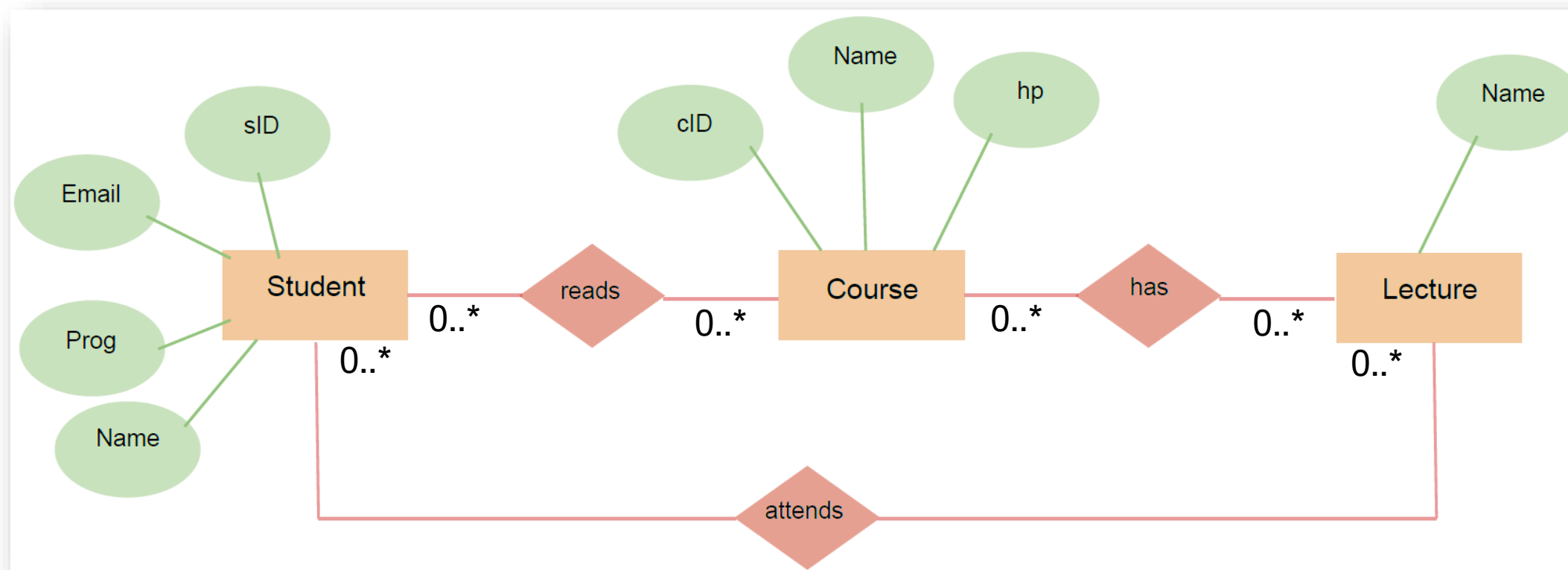
- There can have three kinds of relationship in a conceptual model:
 - **One-to-one**
 - **One-to-many**
 - **Many-to-many**



- To **identify** them, consider only the **MAX** values of the cardinalities

3.2) THE LOGICAL MODEL

- How to go from this drawing to the logical model?



3.2) THE LOGICAL MODEL

- You just follow simple rules:

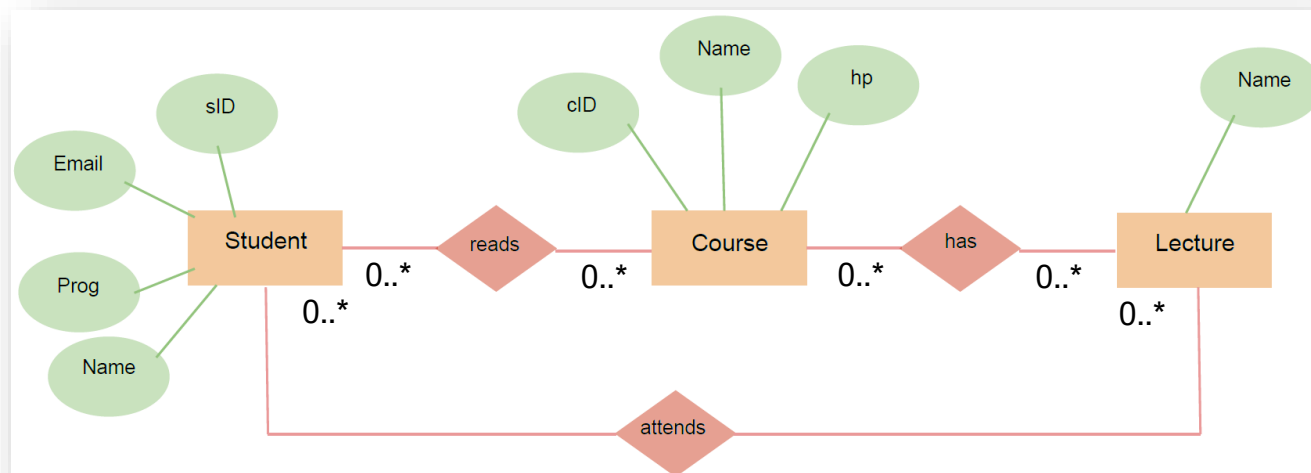
RULES:

- 1) Any **entity** becomes a **table**.
- 2) **Look** at the **cardinalities (max only)**:
 - a. a **one-to-many** cardinality makes the **PK** on the 1 side become a **FK** into the table on the * side.
 - b. a **many-to-many** relation becomes a **full table** with **PK** being the two **FK** of the tables in the relation.

- That is why the cardinalities are **SO IMPORTANT!**

3.2) THE LOGICAL MODEL

- Let's do it!



RULES:

- Any **entity** becomes a **table**.
- Look at the **cardinalities (max only)**:
 - a **one-to-many** cardinality makes the **PK** on the 1 side become a **FK** into the table on the * side.
 - a **many-to-many** relation becomes a **full table** with **PK** being the two **FK** of the tables in the relation.

- TABLES:

3.3) THE PHYSICAL MODEL

- This model deals with the data organization on your hard drive: it is PHYSICAL
- How many files do you need?
- Do you need an index to speed up the search for data?
- How they will be organized in the same directory, in multiple ones, on the same server, on the cloud?...
- For SQLite 3, it is easy, ALL the database will be in ONE physical file only!

3) THE ENTITY-RELATIONSHIP (ER) MODEL

- Once you have done the three (two only with SQLite3) steps: conceptual and logical,
- Congratulations, you have modeled your real-life problem into a computer database!
- Now, we need a **language** to **describe our data** and **interact** with our database

SQL license plate should appear here,
but Sweden does not use "Q" on plates... ☹️

3) THE ENTITY-RELATIONSHIP (ER) MODEL

- By cheating a little...



3) THE ENTITY-RELATIONSHIP (ER) MODEL

- By cheating a little...



3) THE ENTITY-RELATIONSHIP (ER) MODEL

- By cheating a little...



4) STRUCTURED QUERY LANGUAGE (SQL)

- SQL is a language to **Create, Read, Update, and Delete** data from a relational database.

C R U D

- SQL is a "standard" language that works (with several syntax changes) with ALL the different relational databases systems

4) STRUCTURED QUERY LANGUAGE (SQL)

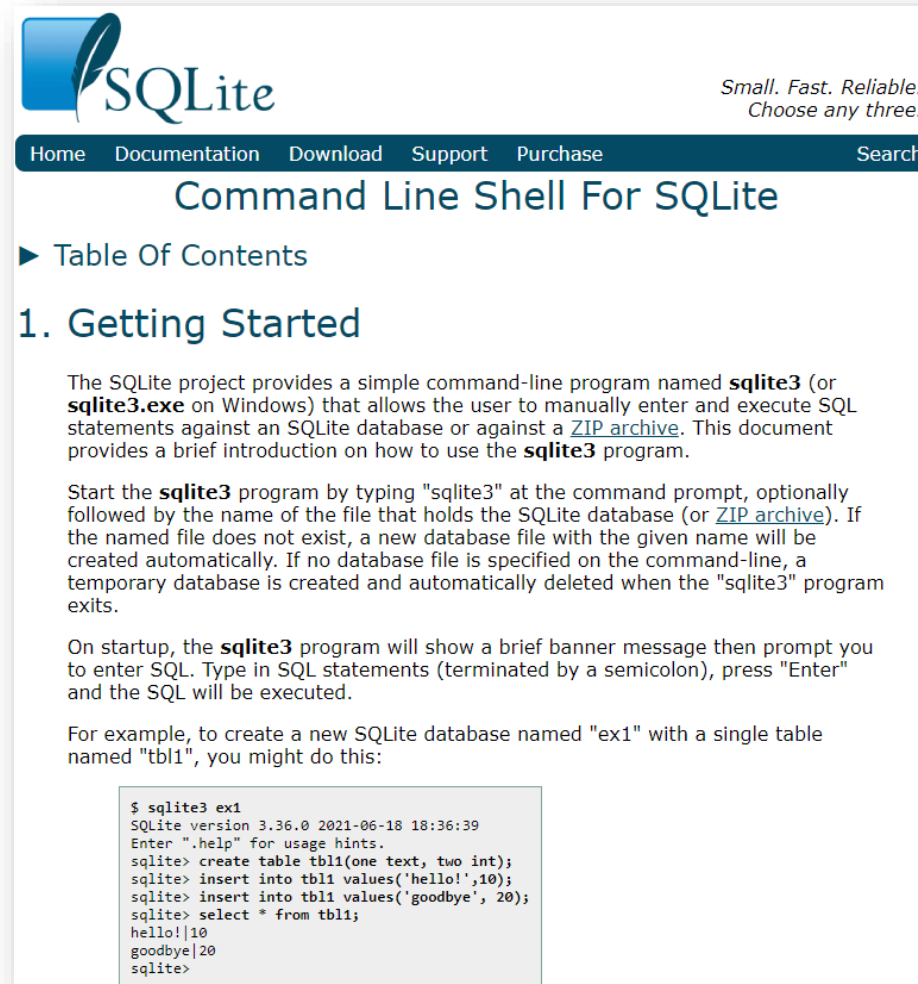
- In this course, we will focus on the SQLite3 syntax but, it can be adapted easily to other DBMS such as Microsoft SQL Server, MySQL, Oracle, ...



Source: <https://www.sqlite.org/>

4) STRUCTURED QUERY LANGUAGE (SQL)

- You can read the "getting started" page to get more information on SQLite3



The screenshot shows the SQLite website's "Command Line Shell For SQLite" page. It features the SQLite logo, a navigation bar with links to Home, Documentation, Download, Support, and Purchase, and a search bar. The main content area includes a "Table Of Contents" section with a link to "1. Getting Started". Below this, there is a paragraph introducing the **sqlite3** program, followed by instructions on how to start the program and a sample command-line session.

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Choose any three.

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Command Line Shell For SQLite

► Table Of Contents

1. Getting Started

The SQLite project provides a simple command-line program named **sqlite3** (or **sqlite3.exe** on Windows) that allows the user to manually enter and execute SQL statements against an SQLite database or against a [ZIP archive](#). This document provides a brief introduction on how to use the **sqlite3** program.

Start the **sqlite3** program by typing "sqlite3" at the command prompt, optionally followed by the name of the file that holds the SQLite database (or [ZIP archive](#)). If the named file does not exist, a new database file with the given name will be created automatically. If no database file is specified on the command-line, a temporary database is created and automatically deleted when the "sqlite3" program exits.

On startup, the **sqlite3** program will show a brief banner message then prompt you to enter SQL. Type in SQL statements (terminated by a semicolon), press "Enter" and the SQL will be executed.

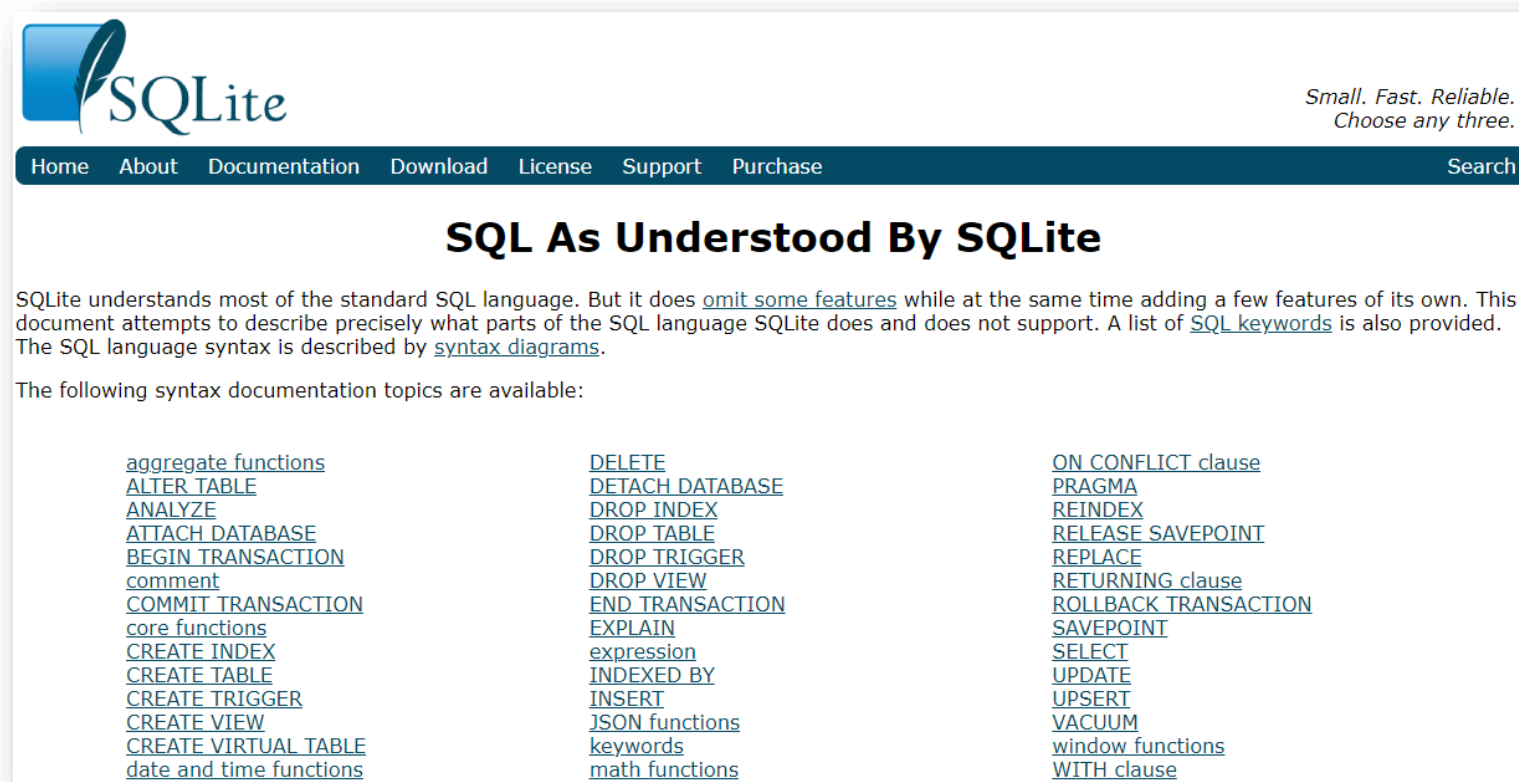
For example, to create a new SQLite database named "ex1" with a single table named "tbl1", you might do this:

```
$ sqlite3 ex1
SQLite version 3.36.0 2021-06-18 18:36:39
Enter ".help" for usage hints.
sqlite> create table tbl1(one text, two int);
sqlite> insert into tbl1 values('hello!',10);
sqlite> insert into tbl1 values('goodbye', 20);
sqlite> select * from tbl1;
hello!|10
goodbye|20
sqlite>
```

Source: <https://www.sqlite.org/cli.html>

4) STRUCTURED QUERY LANGUAGE (SQL)

- The SQLite3 SQL syntax is described on the official site



The screenshot shows the SQLite website. At the top left is the SQLite logo. To the right of the logo is the tagline "Small. Fast. Reliable. Choose any three." Below the logo is a navigation bar with links: Home, About, Documentation, Download, License, Support, Purchase, and a Search bar. The main heading is "SQL As Understood By SQLite". Below this heading is a paragraph explaining that SQLite understands most of the standard SQL language but omits some features and adds its own. It mentions a list of SQL keywords and syntax diagrams. Below the paragraph is a list of syntax documentation topics arranged in three columns.

SQLite *Small. Fast. Reliable. Choose any three.*

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SQL As Understood By SQLite

SQLite understands most of the standard SQL language. But it does [omit some features](#) while at the same time adding a few features of its own. This document attempts to describe precisely what parts of the SQL language SQLite does and does not support. A list of [SQL keywords](#) is also provided. The SQL language syntax is described by [syntax diagrams](#).

The following syntax documentation topics are available:

aggregate functions	DELETE	ON CONFLICT clause
ALTER TABLE	DETACH DATABASE	PRAGMA
ANALYZE	DROP INDEX	REINDEX
ATTACH DATABASE	DROP TABLE	RELEASE SAVEPOINT
BEGIN TRANSACTION	DROP TRIGGER	REPLACE
comment	DROP VIEW	RETURNING clause
COMMIT TRANSACTION	END TRANSACTION	ROLLBACK TRANSACTION
core functions	EXPLAIN	SAVEPOINT
CREATE INDEX	expression	SELECT
CREATE TABLE	INDEXED BY	UPDATE
CREATE TRIGGER	INSERT	UPSERT
CREATE VIEW	JSON functions	VACUUM
CREATE VIRTUAL TABLE	keywords	window functions
date and time functions	math functions	WITH clause

5) SQLITE3

- **SQLite3** is a simple database management system where the data are stored in a single unique **file**

<https://www.sqlite.org/index.html>

- In SQLite3, you can **Create, Read, Update and Delete (CRUD)** information

<https://www.sqlitetutorial.net/>

5) SQLITE3

More information: <https://www.sqlite.org/cli.html>,
<https://www.sqlite.org/lang.html>

- Create:
 - Create a table: CREATE TABLE ...
 - Create new data in a table: INSERT INTO ...
- Read: SELECT * FROM ...
- Update:
 - Update a table content: ALTER TABLE ... (**Be careful**)
 - UPDATE ... SET ... **WHERE** ...
- Delete:
 - Delete a table: DROP TABLE ... (**Be careful**)
 - Delete one or more records (rows) in a table: DELETE FROM ... **WHERE** ...

6) SQLITE3, NODE AND EXPRESS

- This part will be covered in the "Databases" workshops 1 & 2...

ANY QUESTIONS?